PHY-102 Applied Physics
DE-48 (EE-A & B & C)
Assignment #1
CLO-1 Mechanics

Group Assignment

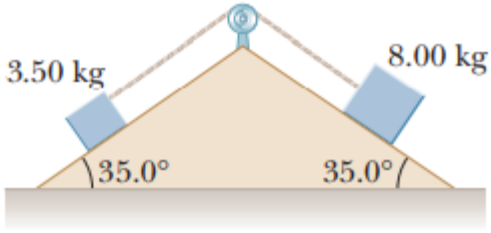
Due Date: Friday 16th October 2026

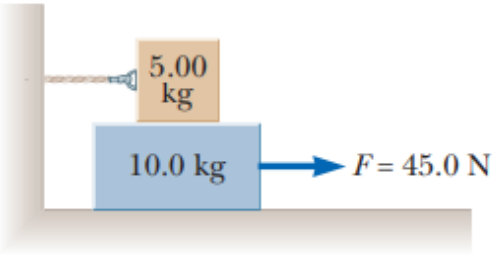
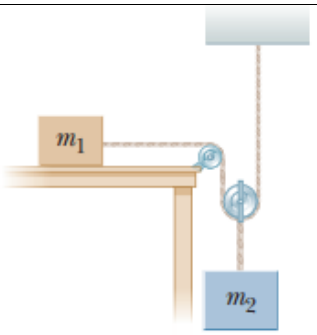
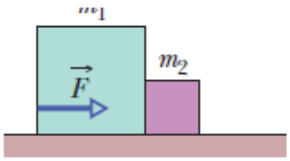
- Answer all questions. Solve each problem according to the strategy discussed in class
- Show complete mathematical work and clearly state assumptions.
- Copied assignments will receive zero marks.
- Draw additional free-body diagrams where required by the solution.
- Use SI units and appropriate significant figures.
- Use g = 9.81 m/s² unless otherwise specified.
- **All assignments must be completed in groups of 4–7 students. Each group member must individually solve at least two questions.**
- **Every student should write name and Reg. No. on each sheet of their work.**

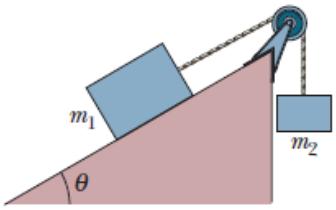
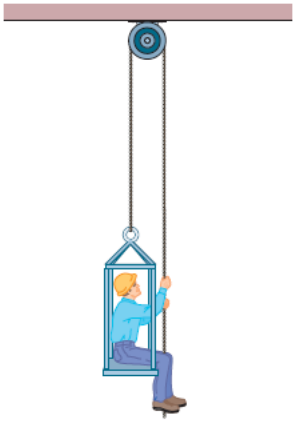
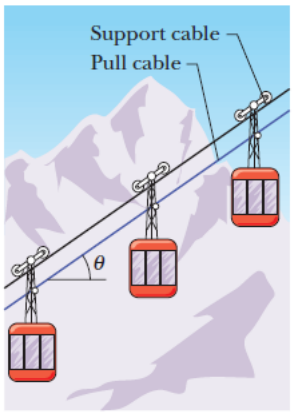
Grading Policy

- thoroughness
- Timely Submission
- Neatness
- Group Presentation

(CLO-1 ⟶ PLO-1)

Part 1 Applications of Newton Laws		
Q-1	Two blocks of mass 3.50 kg and 8.00 kg are connected by a massless string that passes over a frictionless pulley (Fig. P5.49). The inclines are frictionless. a) Find the magnitude of the acceleration of each block. b) the tension in the string. c) The system shown in Figure P5.49 has an acceleration of magnitude 1.50 m/s². Assume that the coefficient of kinetic friction between block and incline is the same for both inclines. Find the coefficient of kinetic friction and the tension in the string.	 **Figure P5.49** Problems 49 and 71.

Q-2	A 5.00-kg block is placed on top of a 10.0-kg block (Fig. P5.70). A horizontal force of 45.0 N is applied to the 10-kg block, and the 5.00-kg block is tied to the wall. The coefficient of kinetic friction between all moving surfaces is 0.200. (a) Draw a free-body diagram for each block and identify the action–reaction forces between the blocks. (b) Determine the tension in the string and the magnitude of the acceleration of the 10.0-kg block.	 Figure P5.70
Q-3	In Figure P5.92, the pulleys and the cord are light, all surfaces are frictionless, and the cord does not stretch. (a) How does the acceleration of block 1 compare with the acceleration of block 2? Explain your reasoning. (b) The mass of block 2 is 1.30 kg. Find its acceleration as it depends on the mass m_1 of block 1. (c) What If? What does the result of part (b) predict if m_1 is very much less than 1.30 kg? (d) What does the result of part (b) predict if m1 approaches infinity? (e) In this last case, what is the tension in the cord? (f) Could you anticipate the answers to parts (c), (d), and (e) without first doing part (b)? Explain.	 Figure P5.92
Q-4	Two blocks are in contact on a frictionless table. A horizontal force is applied to the larger block, as shown in Fig. 5-50. (a) If $m_1 = 2.3$ kg, $m_2 = 1.2$ kg, and F = 3.2 N, find the magnitude of the force between the two blocks. (b) Show that if a force of the same magnitude F is applied to the smaller block but in the opposite direction, the magnitude of the force between the blocks is 2.1 N, which is not the same value calculated in (a). (c) Explain the difference.	 **Figure 5-50** Problem 55.

Q-5	A block of mass $m_1 = 3.70$ kg on a frictionless plane inclined at angle $\theta = 30.0°$ is connected by a cord over a massless, frictionless pulley to a second block of mass $m_2 = 2.30$ kg (Fig. 5-52). What are (a) the magnitude of the acceleration of each block, (b) the direction of the acceleration of the hanging block, and (c) the tension in the cord?	 Figure 5-52 Problem 57.
Q-6	Figure 5-53 shows a man sitting in a bosun's chair that dangles from a massless rope, which runs over a massless, frictionless pulley and back down to the man's hand. The combined mass of man and chair is 95.0 kg. With what force magnitude must the man pull on the rope if he is to rise (a) with a constant velocity and (b) with an upward acceleration of 1.30 m/s²? If the rope on the right extends to the ground and is pulled by a co-worker, with what force magnitude must the co-worker pull for the man to rise (c) with a constant velocity and (d) with an upward acceleration of 1.30 m/s²? What is the magnitude of the force on the ceiling from the pulley system in (e) part a, (f) part b, (g) part c, and (h) part d?	 Figure 5-53 Problem 58.
Q-7	Figure 5-57 shows a section of a cable-car system. The maximum permissible mass of each car with occupants is 2800 kg. The cars, riding on a support cable, are pulled by a second cable attached to the support tower on each car. Assume that the cables are taut and inclined at angle $\theta = 35°$. What is the difference in tension between adjacent sections of pull cable if the cars are at the maximum permissible mass and are being accelerated up the incline at 0.81 m/s²?	 Figure 5-57 Problem 66.

Q-8

In Fig. 6-49, a 49 kg rock climber is climbing a "chimney." The coefficient of static friction between her shoes and the rock is 1.2; between her back and the rock is 0.80. She has reduced her push against the rock until her back and her shoes are on the verge of slipping.
(a) Draw a free-body diagram of her.
(b) What is the magnitude of her push against the rock?
(c) What fraction of her weight is supported by the frictional force on her shoes?



Figure 6-49 Problem 63.

Q-9

a) Derive the expression for velocity of a car moving along a circular banked curve of radius R and calculate the angle of inclination for banked curve in the case of specified velocity without considering frictional force.
b) Find the maximum speed of a car of mass *m* traveling along a banked curve have in order to make the curve without sliding up the incline.
c) Derive an expression for minimum speed the car must have when you include friction or it will start to slide down the incline.

Q-10

An amusement park ride consists of a large vertical cylinder that spins about its axis fast enough that any person inside is held up against the wall when the floor drops away (Fig. P6.59). The coefficient of static friction between person and wall is ms, and the radius of the cylinder is R. (a) Show that the maximum period of revolution necessary to keep the person from falling is $T=\left(\frac{4\pi^2 R\mu_s}{g}\right)^{1/2}$.
(b) If the rate of revolution of the cylinder is made to be somewhat larger, what happens to the magnitude of each one of the forces acting on the person? What happens in the motion of the person?
(c) If the rate of revolution of the cylinder is instead made to be somewhat smaller, what happens to the magnitude of each one of the forces acting on the person? How does the motion of the person change?

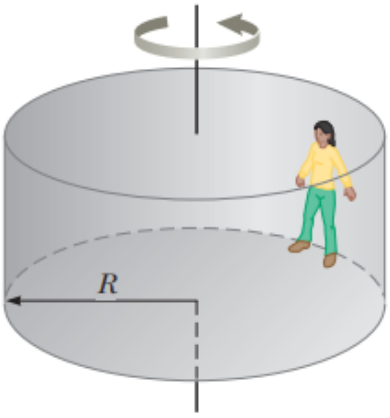


Figure P6.59

Part 2 Work Energy Theorem

Q-1

In Fig. 7-10a, a block of mass m lies on a horizontal frictionless surface and is attached to one end of a horizontal spring (spring constant k) whose other end is fixed. The block is initially at rest at the position where the spring is unstretched (x = 0) when a constant horizontal force in the positive direction of the x axis is applied to it. A plot of the resulting kinetic energy of the block versus its position x is shown in Fig. 7-36. The scale of the figure's vertical axis is set by K_s = 4.0 J. (a) What is the magnitude of F? (b) What is the value of k?

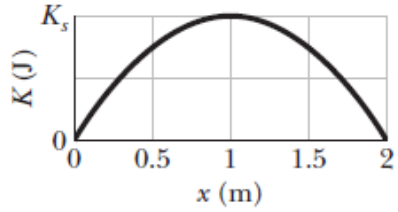
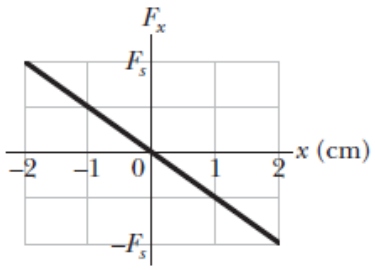
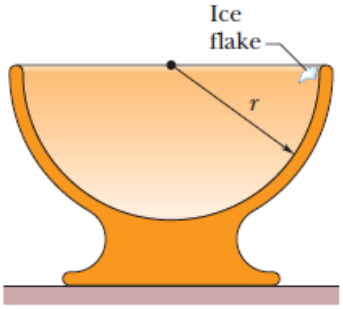
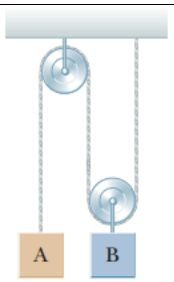
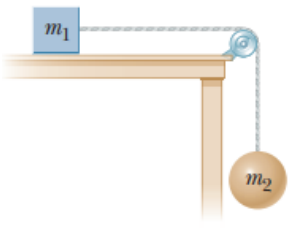
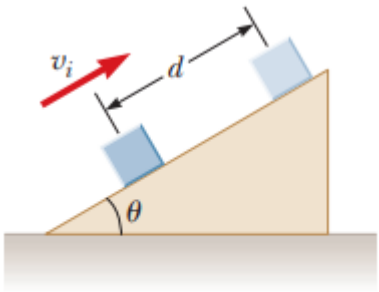
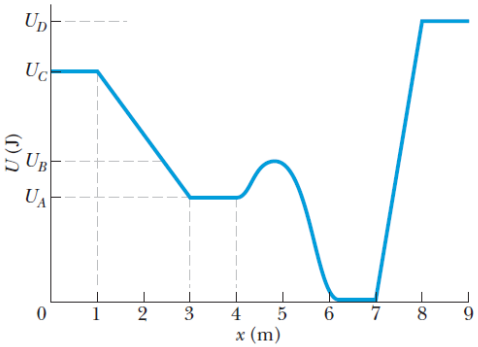
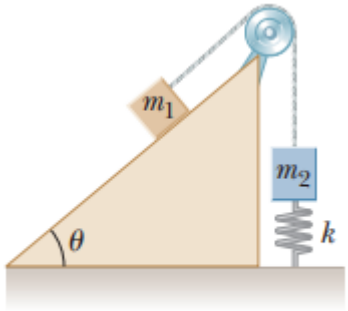
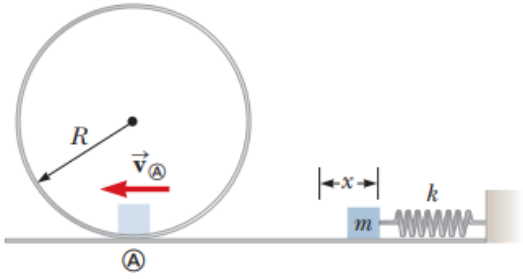


Figure 7-36 Problem 30.

Q-2	Figure 7-37 gives spring force F_x versus position x for the spring–block arrangement of Fig. 7-10. The scale is set by F_s = 160.0 N. We release the block at x = 12 cm. How much work does the spring do on the block when the block moves from x_i = +8.0 cm to (a) x = +5.0 cm, (b) x = −5.0 cm, (c) x = −8.0 cm, and (d) x = −10.0 cm?	 **Figure 7-37** Problem 32.
Q-3	In Fig. 8-32, a 2.00 g ice flake is released from the edge of a hemispherical bowl whose radius r is 22.0 cm. The flake–bowl contact is frictionless. (a) How much work is done on the flake by the gravitational force during the flake's descent to the bottom of the bowl? (b) What is the change in the potential energy of the flake–Earth system during that descent? (c) If that potential energy is taken to be zero at the bottom of the bowl, what is its value when the flake is released? (d) If, instead, the potential energy is taken to be zero at the release point, what is its value when the flake reaches the bottom of the bowl? (e) If the mass of the flake were doubled, would the magnitudes of the answers to (a) through (d) increase, decrease, or remain the same?	 **Figure 8-32** Problems 5 and 11.
Q-4	The system shown in Figure P8.11 consists of a light, inextensible cord, light, frictionless pulleys, and blocks of equal mass. Notice that block B is attached to one of the pulleys. The system is initially held at rest so that the blocks are at the same height above the ground. The blocksare then released. Find the speed of block A at the moment the vertical separation of the blocks is *h*.	 **Figure P8.11**
Q-5	The coefficient of friction between the block of mass m_1=3.00 kg and the surface in Figure P8.22 is μ_K=0.400. The system starts from rest. What is the speed of the ball of mass m_2 =5.00 kg when it has fallen a distance h =1.50 m?	 **Figure P8.22**

Q-6	A 5.00-kg block is set into motion up an inclined plane with an initial speed of v_i=8.00 m/s (Fig. P8.23). The block comes to rest after traveling d=3.00 m along the plane, which is inclined at an angle of θ=30.08 to the horizontal. For this motion, determine (a) the change in the block's kinetic energy, (b)the change in the potential energy of the block–Earth system. (c) the friction force exerted on the block (assumed to be constant). (d) What is the coefficient of kinetic friction?	 Figure P8.23
Q-7	Tarzan, who weighs 688 N, swings from a cliff at the end of a vine 18 m long (Fig. 8-40). From the top of the cliff to the bottom of the swing, he descends by 3.2 m. The vine will break if the force on it exceeds 950 N. (a) Does the vine break? (b) If no, what is the greatest force on it during the swing? If yes, at what angle with the vertical does it break?	 Figure 8-40 Problem 27.
Q-8	Figure 8-49 shows a plot of potential energy U versus position x of a 0.200 kg particle that can travel only along an x axis under the influence of a conservative force. The graph has the values U_A = 9.00 J, U_C = 20.00 J, and U_D = 24.00 J. The particle is released at the point where U forms a "potential hill" of "height" U_B = 12.00 J, with kinetic energy 4.00 J. What is the speed of the particle at (a) x = 3.5 m and (b) x = 6.5 m? What is the position of the turning point on (c) the right side and (d) the left side?	 Figure 8-49 Problem 38.
Q-9	A block of mass m_1=20.0 kg is connected to a block of mass m_2=30.0 kg by a massless string that passes over a light, frictionless pulley. The 30.0-kg block is connected to a spring that has negligible mass and a force constant of k=250 N/m as shown in Figure P8.64. The spring is unstretched when the system is as shown in the figure, and the incline is frictionless. The 20.0-kg block is pulled a distance h=20.0 cm down the incline of angle θ=40.08 and released from rest. Find the speed of each block when the spring is again unstretched.	 Figure P8.64

Q-10	A block of mass 0.500 kg is pushed against a horizontal spring of negligible mass until the spring is compressed a distance x (Fig. P8.65). The force constant of the spring is 450 N/m. When it is released, the block travels along a frictionless, horizontal surface to point A, the bottom of a vertical circular track of radius R =1.00 m, and continues to move up the track. The block's speed at the bottom of the track is v_A=12.0 m/s, and the block experiences an average friction force of 7.00 N while sliding up the track. (a) What is x? (b) If the block were to reach the top of the track, what would be its speed at that point? (c) Does the block actually reach the top of the track, or does it fall off before reaching the top?	 Figure P8.65
